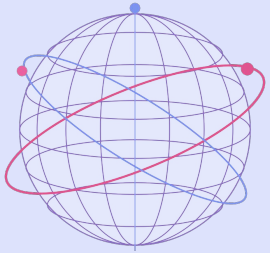
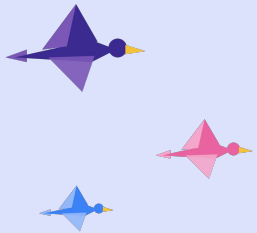
PARTICIPANT PRESENTATION

Qiskit Fall Fest 2026

Hackathon · Team Details





Participant Details

PROJECT TITLE
Q-Resilience AI · Brand: Q-Silience

TRACK
Quantum + AI & ML

TEAM NAME
Ctrl_AI

TEAM LEADER
Dhyan P

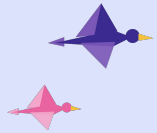
EMAIL / CONTACT
dhyanp1355@gmail.com

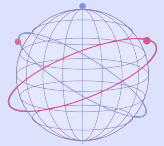
TEAM MEMBERS
Dhyan P · Sangamesh Athani · Harsha Hiregoudar

INSTITUTION / ORGANISATION
Jain College of Engineering and Research, Belagavi

COUNTRY
India

Problem Statement





THE PROBLEM

Microgrids juggle demand, solar, battery, voltage and frequency at the same time. Single-threshold alarms watch one signal at a time and can miss unusual combinations, such as a demand surge while the battery is low and solar output drops.

RESEARCH QUESTION

Can a quantum-kernel classifier (QSVC) label simulated microgrid records as normal or anomalous, and how does it compare with a classical SVC on the same held-out data?

WHY IT MATTERS

1. Demand surges, solar drops, low battery and voltage or frequency drift often combine to threaten stable operation.
2. Q-Silience, our dashboard, makes model outputs and sensor patterns easy for an operator to inspect.
3. A fair quantum-vs-classical comparison on identical data turns a hype question into a measured result.

Objectives & Novelty

OBJECTIVES

1. Generate or load simulated microgrid sensor records: demand, solar, battery SOC, voltage, frequency
2. Train a Qiskit quantum-kernel QSVC and a scikit-learn SVC on the same split
3. Evaluate both on held-out data: accuracy, precision, recall, F1, runtime
4. Present predictions, metrics and scenarios in the Q-Silience dashboard

NOVELTY

- **Measurable experiment.** One protocol scores quantum and classical models on the same held-out set.
- **Scenario Simulation Lab.** Each scenario changes real inputs sent to the backend; no hardcoded labels.
- **Transparent by design.** Readiness indicators, error states, simulation labels and stated limits.

Solution & Quantum Advantage

Sensor data › Preprocessing › **QSVC ‖ SVC** › Held-out evaluation › Dashboard

THE SOLUTION

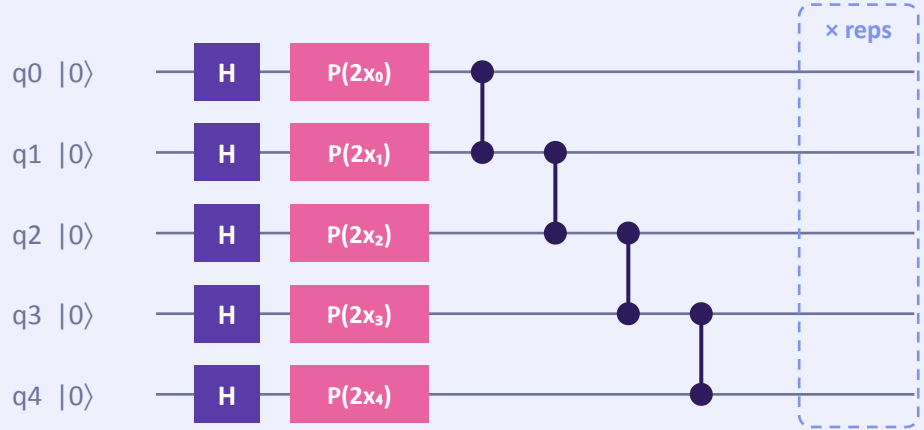
- ZZFeatureMap encodes scaled sensor features into quantum states
- A fidelity-based quantum kernel feeds a Qiskit QSVC
- A scikit-learn SVC runs as the baseline on identical data
- Dashboard shows the prediction and which model produced it

QUANTUM ADVANTAGE: OUR POSITION

- We do not claim quantum advantage
- We measure whether a quantum kernel is competitive: accuracy, F1 and runtime on the same held-out data
- Recorded run: classical SVC led (F1 0.90 vs 0.74). Simulator only, so no hardware claim

Algorithm & Circuit Design

Illustrative ZZFeatureMap schematic: 5 features → 5 qubits



H = Hadamard · P = phase rotation by feature x_i · ●—● = entangling interaction

KEY DETAILS

- **Encoding:** features scaled to [0, 0.5], then ZZFeatureMap (reps = 1)
- **Kernel:** fidelity-based, exact statevector
- **Classifier:** QSVC from Qiskit Machine Learning
- **Execution:** simulator unless hardware is explicitly configured
- **Logged:** 5 qubits, linear entanglement, kernel method, runtime

Implementation & Tech Stack

FRONTEND*
React · TypeScript · Vite · charts

BACKEND API
Python · FastAPI · Uvicorn

QUANTUM
Qiskit + Qiskit Machine Learning (QSVC, simulator)

CLASSICAL ML
scikit-learn SVC, scaling, metrics

DATA
NumPy · pandas · synthetic generator

HOSTING
Vercel (frontend) · Render Web Service (API)

PROPOSED API ENDPOINTS

`GET /health`
API reachable

`GET /ready`
model readiness

`POST /api/predict`
one sensor record

`POST /api/evaluate`
held-out metrics

`POST /api/scenario`
scenario + predictions

**Likely stack: confirm against package.json. Endpoints are the proposed design.*

Results & Analysis

Quantum vs classical on the held-out test set (75 records)

Model	Accuracy	Precision	Recall	F1	Runtime
QSVC (quantum)	0.867	0.875	0.636	0.737	1.3 s
SVC (classical)	0.947	0.950	0.864	0.905	<0.01 s

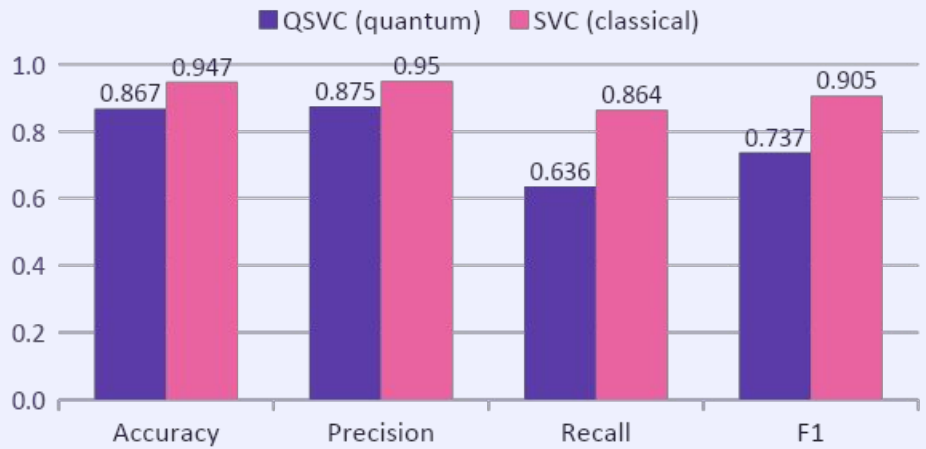
Recorded run · synthetic data. SVC led: 94.7% vs 86.7% accuracy; QSVC missed 8 of 22 anomalies vs 3 for SVC. Reported as measured; no quantum advantage claimed.

EVALUATION PROTOCOL

- Same split and scaling for both models; scaler fit on training data only
- QSVC feature scale and reps chosen on a validation split, never the test set
- Report accuracy, precision, recall, F1, confusion matrix, runtime
- Recorded: seed 42, 300 records, 25% test, 30% anomalous
- Claim superiority only if the measurements show it

Where Each Model Fails

Held-out metrics, QSVC vs SVC (75 test records)

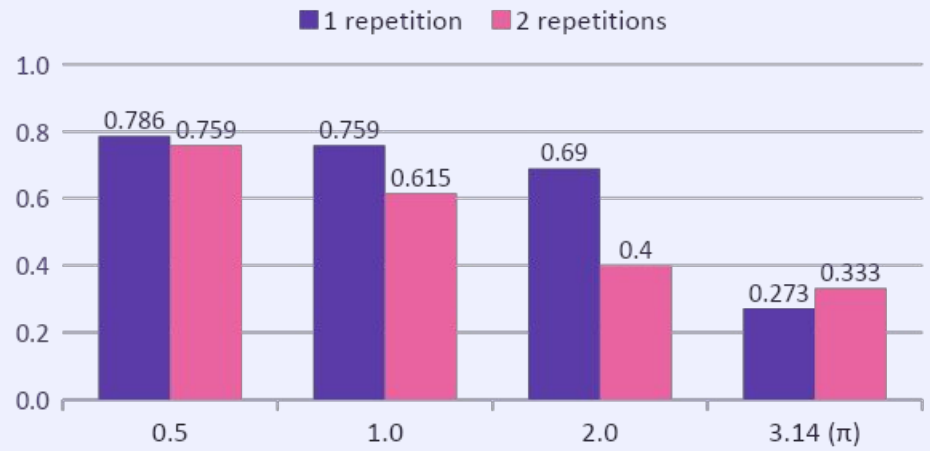


ERRORS ON 75 TEST RECORDS

- QSVC: caught 14 of 22 anomalies, 2 false alarms
- SVC: caught 19 of 22 anomalies, 1 false alarm
- Recall is the gap: in a grid, a missed anomaly costs more than a false alarm
- Both models rarely cry wolf (precision 0.88 vs 0.95)

Insight: Encoding Scale Matters

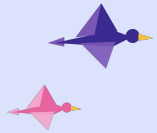
Validation F1 by feature scale (training data only)

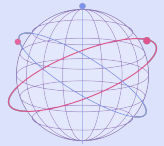


WHAT WE LEARNED

- Same circuit, same data: F1 swung from 0.79 to 0.27 on feature scale alone
- Small rotations and one repetition worked best
- Deeper circuits scored lower at 3 of 4 scales
- Classical RBF SVC needed no tuning to beat it
- Small validation set (57 records): indicative, not final

Impact & Real-World Applications





1

Microgrid monitoring

Surfaces unusual combinations of demand, solar, battery, voltage and frequency that single-threshold checks can miss. Predictions are model outputs, not confirmed faults.

2

Measurable QML research

A fair comparison under one protocol. On our run the classical SVC led, and we report that openly.

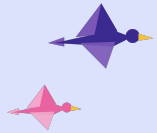
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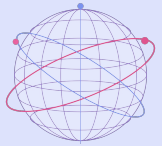
Reusable demo pipeline

An end-to-end template covering data, preprocessing, QSVC, SVC, API and dashboard for teaching or extension.

Prototype on simulated data: not for controlling grid equipment or safety-critical decisions.

Challenges & Future Scope





CHALLENGES

- Synthetic labels may encode the generation rules, so high scores do not prove real-world performance
- Simulator results are not the same as running on quantum hardware
- QSVC trailed SVC on accuracy (86.7% vs 94.7%) and runtime on this small dataset
- Free hosting means cold starts and limits; pin and test Python versions

FUTURE SCOPE

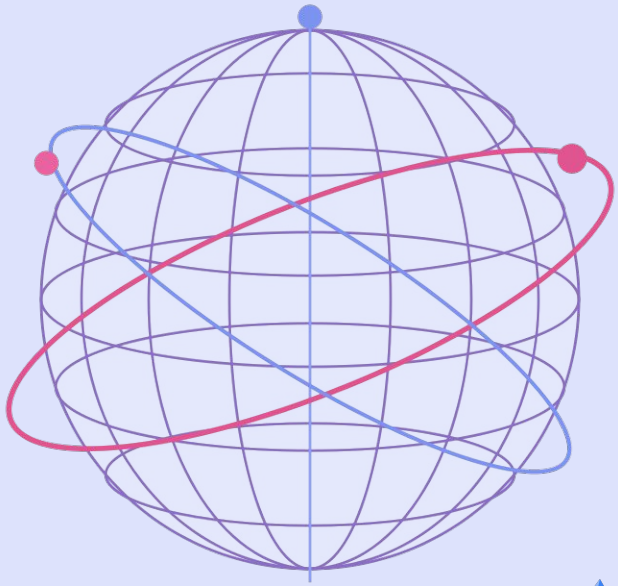
- Validate on real or public microgrid datasets
- Run on quantum hardware through Qiskit Runtime once configured
- Add a Quantum Circuit Explorer and Anomaly Investigation Center for interpretability
- Add microgrid digital-twin visuals and event history / export

CLOSING

Thank You

Questions?

Inspiring Minds. Shaping Tomorrow.



GITHUB REPOSITORY
https://github.com/vtu-qiskit-fall-fest-2026/vtu-hc-ctrl-ai.git

DEMO VIDEO LINK
[https://youtu.be/HAgrzv3rs3k]